

Solutions – Quiz 4

ECON 308 – International Economic Relations

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Multiple Choice (1 mark each)

1. *If a country's terms of trade improve, its welfare will:*

Correct: A. *Always increase.* In the standard (distortion-free) model, an improvement in the terms of trade (export price relative to import price) pivots the trade (isovalue) line outward around the production point, allowing the country to reach a higher indifference curve.

2. *Growth biased toward a country's export good tends to:*

Correct: B. *Deteriorate its terms of trade.* Export-biased growth expands the world supply of the export good and tends to lower its relative price, worsening the home country's terms of trade (for a price-taking small country, prices are exogenous, but the sign remains the same in general equilibrium intuition).

3. *If the world relative price of the country's export good rises due to global demand:*

Correct: A. *Improve TOT and raise welfare.* A higher relative price of exports is an improvement in the TOT, shifting the consumption possibility outward and increasing welfare.

4. *If the world relative price of cloth rises and the country exports cloth, it will:*

Correct: B. *Experience an improvement in its terms of trade.* The export good becomes relatively more valuable, raising real income at given production.

5. *In the Standard Trade Model, changes in welfare are illustrated by:*

Correct: C. *Tangency between the highest indifference curve and the trade line.* Welfare rises when the budget (trade) line at world prices supports a higher indifference curve.

Problem 1 (7 marks)

Data. Goods: food (F) and manufactures (M). PPF: $F^2 + M^2 = 100$. World relative price: $P_M/P_F = 2$. Consumption satisfies $M = 0.5F$.

1) Production point (F_P, M_P) at given prices. With F on the horizontal axis and M on the vertical, the PPF slope is the marginal rate of transformation (MRT):

$$\text{PPF: } F^2 + M^2 = 100 \Rightarrow 2F dF + 2M dM = 0 \Rightarrow \frac{dM}{dF} = -\frac{F}{M}.$$

The isovalue (trade) line has slope $-\frac{P_F}{P_M}$. Tangency (value-maximizing production) requires

$$-\frac{F}{M} = -\frac{P_F}{P_M} \Rightarrow \frac{F}{M} = \frac{P_F}{P_M} = \frac{1}{2} \Rightarrow F = \frac{M}{2}.$$

Impose on PPF:

$$\left(\frac{M}{2}\right)^2 + M^2 = 100 \Rightarrow \frac{M^2}{4} + M^2 = \frac{5}{4}M^2 = 100 \Rightarrow M^2 = 80 \Rightarrow M_P = \sqrt{80} = 4\sqrt{5} \approx 8.944.$$

Hence $F_P = \frac{M_P}{2} \approx 4.472$.

$$(F_P, M_P) \approx (4.472, 8.944).$$

2) Consumption point (F_C, M_C) given $M = 0.5F$ and trade at world prices. Normalize prices by $P_F = 1$, $P_M = 2$ (only relative prices matter). The value of production (national income at world prices) is

$$V = P_F F_P + P_M M_P = 1 \cdot 4.472 + 2 \cdot 8.944 \approx 4.472 + 17.888 = 22.36.$$

Budget (trade) line at world prices:

$$P_F F + P_M M = 22.36 \Rightarrow F + 2M = 22.36.$$

Consumption must lie on the ray $M = 0.5F$. Substitute into the budget:

$$F + 2(0.5F) = 2F = 22.36 \Rightarrow F_C = 11.18, \quad M_C = 0.5F_C = 5.59.$$

$$(F_C, M_C) \approx (11.18, 5.59).$$

Trade volumes: Exports of M : $M_P - M_C \approx 8.944 - 5.59 = 3.354$. Imports of F : $F_C - F_P \approx 11.18 - 4.472 = 6.708$. Valued at world prices, export revenue = $2 \times 3.354 = 6.708$ equals import expenditure = 1×6.708 (trade is balanced).

3) Terms of trade and gains from trade. The country's terms of trade are $P_M/P_F = 2$ (given). Production is chosen to maximize the value at these prices; then trading at the same prices allows the country to move from (F_P, M_P) on the PPF to (F_C, M_C) on a *higher* indifference curve (the trade line through the production point supports a strictly higher utility than autarky, unless preferences happen to make the autarky point coincidentally optimal at those prices). Hence, under standard assumptions:

$$\text{Welfare rises with trade at } P_M/P_F = 2.$$

Problem 2 (8 marks)

Data. Before growth: $Q_T = 200 - 2Q_C$ (textiles T , chemicals C), world price ratio $P_T/P_C = 1$. After biased growth toward T : $Q_T = 240 - 2Q_C$.

1) Change in maximum production possibilities (intercepts). Intercepts are where the other good is zero.

Before growth: If $Q_C = 0$, $Q_T^{\max} = 200$; if $Q_T = 0$, $Q_C^{\max} = 100$.

After growth: If $Q_C = 0$, $Q_T^{\max} = 240$; if $Q_T = 0$, $Q_C^{\max} = 120$.

So

$$\Delta Q_T^{\max} = +40, \quad \Delta Q_C^{\max} = +20.$$

Growth is T -biased (larger outward shift in the T -intercept).

2) Change in export potential at fixed world prices ($P_T = P_C = 1$). At prices $(1, 1)$, the isovalue line has slope -1 while the PPF slope is constant at -2 . Value maximization chooses the corner with the flatter price line: set $Q_C = 0$ (produce only T).

Before: $(Q_T, Q_C) = (200, 0) \Rightarrow \text{income } V = 200$. After: $(Q_T, Q_C) = (240, 0) \Rightarrow V = 240$.

With homothetic preferences (standard in the STM), expenditure shares depend only on prices. Let s_T be the consumption share of T at $(1, 1)$. Then exports of T equal production minus consumption:

$$X_T = Q_T - \frac{s_T V}{P_T} = (1 - s_T)Q_T$$

(since $V = P_T Q_T + P_C Q_C = Q_T$ at $Q_C = 0$ and $P_T = P_C = 1$). Therefore

$$X_T^{\text{after}} = (1 - s_T) \cdot 240 = \frac{240}{200} X_T^{\text{before}}.$$

Export potential of T rises by 20% at the given prices.

3) Effects on terms of trade (TOT) and welfare.

- **Small-country case (prices fixed at world levels):** Income rises from 200 to 240 and the consumption set expands; welfare *unambiguously increases*. Terms of trade are unchanged by assumption.
- **Large-country case (endogenous prices):** T -biased growth increases the world supply of T , tending to *lower* P_T/P_C (a TOT *deterioration*). Welfare depends on the balance between higher productive capacity and worse TOT. Typically welfare still rises; only in extreme cases with very elastic foreign demand could TOT deteriorate so much that welfare falls (*immiserizing growth*).

T -biased growth tends to worsen the home TOT.

Welfare rises unless the TOT deterioration is exceptionally large.